

What can a shore camera actually see?

Detect / recognize / identify coverage of a 30 m EO/IR camera at the port of Esbjerg, Denmark, tested against one week of real AIS traffic (1.7 M positions, 159 vessels, 13-19 July 2026).

59%

of moving-vessel time the camera recognizes the vessel (51-66% across height assumptions)

89%

of vessel transits are recognized at least once

16% / 90%

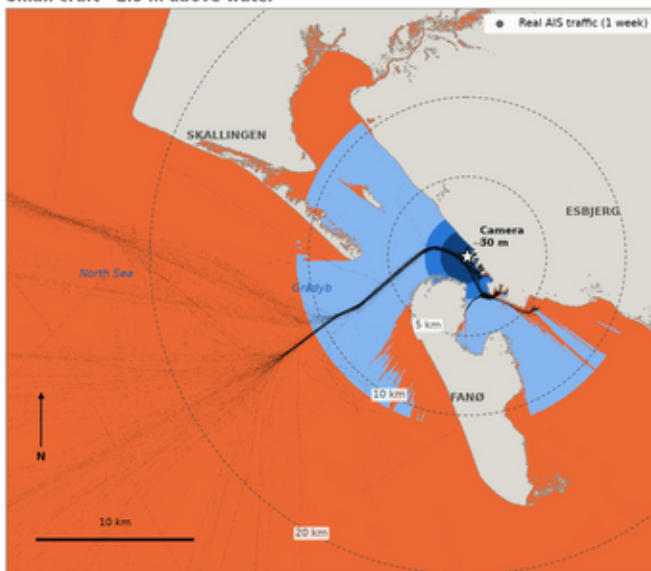
pleasure craft vs ferries recognized. Pleasure craft not seen at all 53% of the time

+12 pts

from a second camera at the best of 77 tested coastal sites (59% to 71%)

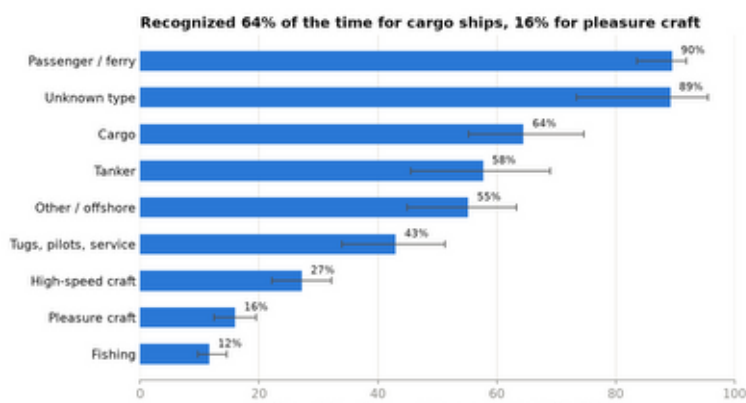
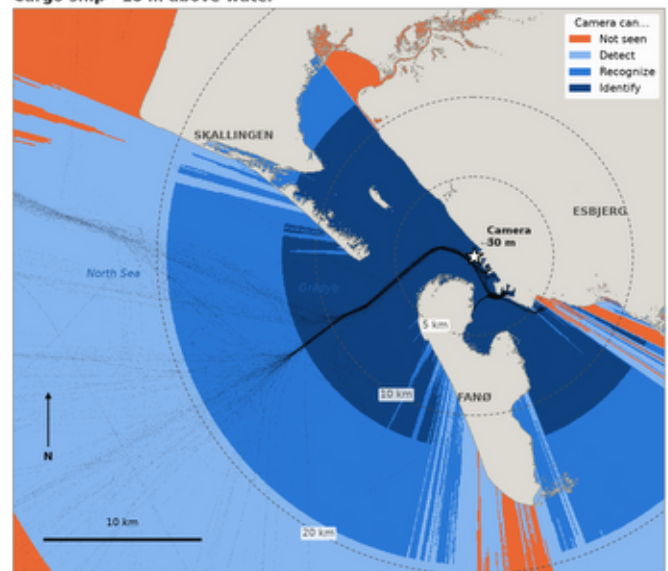
What can a shore camera actually see? Day camera, wide setting (25 mm), mounted 30 m above sea level in Esbjerg

Small craft · 2.5 m above water



AIS: Danish Maritime Authority, 13-19 Jul 2026 | Elevation: Copernicus GLO-30 DEM | Model & analysis: N. Shaprut

Cargo ship · 18 m above water



Bars: central height estimate. Whiskers: vessel heights -30% / +30%. Moving vessels only, best of day camera / thermal.

WHAT IT MEANS FOR AN OPERATOR

- The approach channel is covered: ferries and cargo ships are recognized most of the time, and almost every transit is recognized at least once.
- Small craft are the gap. Pleasure craft are not seen at all 53% of the time, and many small boats carry no AIS, so the real gap is larger.
- Zoom lifts recognition from 55% to 81%, but through a 3° field of view. Range versus coverage is the core design choice.
- A second camera at the Skallingen tip adds +12 pts overall (+5.5 for small craft). All 77 candidate sites are ranked and mapped.

METHOD

AIS cleaned (duplicates, position spikes, moored-boat speed jitter, land transponders) and split into transits. Line of sight along 1,800 rays over the Copernicus GLO-30 elevation model, with Earth curvature and refraction. Visible height becomes pixels on target for three sensors (EO wide 25 mm, EO zoom 100 mm, thermal 50 mm), EO only in daylight. Johnson criteria: 2 / 8 / 12.8 px. Every number is computed with vessel heights -30% and +30%.

VALIDATION & DATA QUALITY

Matches the closed-form curvature result exactly over open sea, and agrees with Google Earth views from the camera position. Findings that changed the rules: 22% duplicate receptions, 14,504 'moving' positions from moored boats, a transponder on a trailer driving through town, offshore-wind vessels hidden as 'Other'. Limits: geometric best case - no haze, glare or sea state.